This framework formalizes the workflow into a "Policy Extraction & Validation Protocol.”

---

1. Structural Discovery: The Outline Heuristic

NDCs are not standardized; they vary by country. To find policy-rich sections, you must look for specific keywords that signal the country’s commitments.

Standard Sections

- **Mitigation:** Actions aimed at reducing greenhouse gas (GHG) emissions (e.g., transitioning to renewables).
- **Adaptation:** Actions aimed at adjusting to the current and future effects of climate change (e.g., sea walls, drought-resistant crops).
- **Means of Implementation (MoI):** Where finance, technology transfer, and capacity-building are discussed.

Non-Standard Indicators (Where to look if the above are missing)

If the outline is vague, look for these "Proxy Sections" in the Table of Contents:

- **Sectoral Planning:** Sections titled "Energy," "Transport," "Agriculture," or "Waste Management."
- **Priority Areas:** Often listed as "Key Pillars" or "Strategic Directions."
- **Cross-Cutting Issues:** Sections on gender, youth, or indigenous peoples often contain localized adaptation policies.
- **Investment Portfolios:** Lists of projects seeking international funding.

---

2. Extraction: The Policy Nomenclature

Policies are rarely labeled "Policy #1." Countries use diverse terminology to describe their commitments. A comprehensive extraction tool should scan for:

- **Action-Oriented:** Action Plan, Action Item, Measure, Initiative, Intervention, Program.
- **Strategic:** Roadmap, Strategy, Long-term Vision, Policy Framework, Master Plan.
- **Regulatory:** Legislative Act, Decree, Mandate, Building Code, Standard (e.g., MEPS - Minimum Energy Performance Standards).
- **Economic:** Fiscal Instrument, Subsidy, Carbon Tax, Feed-in Tariff, Incentive Scheme.
- **Project-Based:** Pilot Project, Flagship Program, Infrastructure Investment.

---

3. The Validation Framework: Valid vs. Non-Sound

The core of your system is distinguishing between "rhetoric" and "commitment." We can define a **Valid Policy** through three dimensions: **Specificity, Tangibility, and Scalability.**

A. Characteristics of a Valid Policy

A policy is valid if it contains **Observable Parameters**.

1. **Quantifiable Targets:** The policy includes numbers, dates, and units.
    - *Example:* "Retrofit 5,000 public buildings for 30% energy efficiency by 2030."
2. **Binding Mechanisms:** The policy is backed by law or regulation, not just "encouragement."
    - *Example:* "Implementing a mandatory 10% biofuel blend for all commercial transport by 2027."
3. **Spatial Specificity:** It identifies exactly *where* the impact will occur.
    - *Example:* "Restoring 500 hectares of mangroves in the [X] River Delta."
4. **Technological Shift:** It specifies a transition from one state to another.
    - *Example:* "Phasing out all coal-fired power plants with a capacity over 100MW by 2040."

B. Characteristics of a Non-Sound Policy

A policy is "non-sound" or "weak" if it is **Performative** or **Vague**.

1. **Process without Outcome:** Focuses on meetings rather than results.
    - *Example:* "Strengthening dialogue with environmental NGOs" or "Promoting awareness of climate change."
2. **Aspirational Language:** Uses verbs like "seek to," "explore," "strive," or "aim" without a roadmap.
    - *Example:* "We aim to explore the possibility of hydrogen power in the future."
3. **Lack of Baseline:** Offers a change without stating the starting point.
    - *Example:* "We will increase green spaces" (without saying by how much or where).
4. **Administrative Maintenance:** Standard government operations rebranded as climate action.
    - *Example:* "Maintaining the existing Ministry of Environment budget."

---

3. District energy systems

- If framed as decarbonizing heating and cooling → **Mitigation**
- If framed as ensuring energy reliability during extreme weather → **Adaptation (Resilience)**
- Often legitimately **both**, but the dominant intent must be identified.

4. Floodable parks and wetlands

- If framed as reducing flood risk through natural absorption → **Nature-Based Solutions + Adaptation**
- If framed as biodiversity enhancement with no climate-risk framing → **Nature-Based Solutions only**

5. Infrastructure hardening

- Strengthening power grids, water systems, transport networks to withstand storms or heat → **Adaptation (Resilience)**
  Even if energy-related, this is not mitigation unless emissions reduction is the main objective.

Resilience should be a tag or attribute, not a standalone class

- Resilience is **not a distinct policy mechanism**
- It describes **how well a system performs under stress**, not what it does
- Treating it as a class leads to overlap and confusion with adaptation

Logic: Use **Adaptation** as the category, and represent resilience as a **subcategory**.

Example

- Category: Adaptation
- Sub-category: Resilience

Policy Type	Valid / High Merit	Non-Sound / Low Merit
Transport	Convert 20k city buses to electric by 2035.	Encourage citizens to use public transport more often.
Urban	Increase urban canopy cover by 2% via mandatory green roofs.	Study the benefits of trees in urban environments.
Fiscal	Implementation of a $25/ton carbon tax on heavy industry.	Support the idea of "polluter pays" in principle.
Building	Update national codes to require Net-Zero status for new builds.	Promote sustainable architecture through workshops.
Energy	Install 5GW of offshore wind capacity by 2030.	Facilitate partnerships for renewable energy research.

---

5. The "Global Consistency" Filter

Valid policies are often **replicable**. A strong policy is one that has a proven "logic of intervention." If a policy like "Building Codes" appears across multiple NDCs, it gains validity because:

- **Implementation exists:** There is a known path to execute it.
- **Benchmarking is possible:** You can compare the stringency of one country's building code against another's.
- **Data availability:** These policies usually produce measurable data (e.g., energy saved per square meter).

Mitigation

Mitigation policies are designed to influence the climate system by changing the amount of greenhouse gases humans release into the atmosphere or by increasing the ability of natural or engineered systems to remove those gases from the atmosphere. The central idea behind mitigation is that human activities such as energy production, transportation, industrial processes, and land use are major contributors to climate change, and that altering these activities can slow or limit future warming.

A policy falls under mitigation when its main purpose is to reduce emissions at their source, avoid future emissions, or actively remove carbon dioxide or other greenhouse gases from the atmosphere. This includes policies that shift energy supply toward low-carbon sources, reduce reliance on fossil fuels, improve the carbon intensity of production systems, or expand carbon sinks through deliberate intervention. Even when mitigation policies deliver additional benefits, such as improved air quality or economic savings, they are classified as mitigation if their primary intent is to affect greenhouse gas concentrations and long-term climate forcing.

Mitigation policies are forward-looking in nature. They are concerned with preventing future climate change rather than responding to climate impacts that have already begun. If a policy's success is measured by emissions reduced, carbon sequestered, or progress toward decarbonization targets, it is a mitigation policy.

Operational Definition

A policy is **Mitigation** if its **primary causal pathway** is:
Human activity → ↓ GHG emissions OR ↑ carbon sinks → ↓ atmospheric forcing

What the model should look for

- Direct or indirect **emissions reduction**
- **Energy system transformation**
- **Decarbonization targets** or trajectories
- **Carbon sequestration** framed in climate terms

Typical policy mechanisms

- Renewable energy mandates
- Electrification (transport, buildings)
- Carbon pricing, offsets, caps
- Fuel switching
- Carbon capture or storage

Disambiguation rule
If the policy **acts on the climate system itself**, not on vulnerability to impacts → **Mitigation**.

Adaptation

Adaptation policies are designed to reduce the negative consequences of climate change by helping people, ecosystems, and infrastructure cope with current or expected climate impacts. Unlike mitigation, adaptation does not attempt to slow climate change itself. Instead, it assumes that climate change is already happening or will happen and focuses on reducing harm. Resilience is a core component of adaptation. It refers to a system's capacity to continue functioning under climate stress, to absorb disturbances without collapsing, and to recover quickly after disruption. A policy that strengthens resilience does so by improving robustness, flexibility, redundancy, preparedness, or recovery capacity in the face of climate hazards such as heatwaves, flooding, droughts, storms, or sea-level rise. A policy should be classified as adaptation when its primary logic begins with a climate risk and ends with reduced exposure, reduced sensitivity, or increased adaptive capacity, resulting in reduced vulnerability, reduced damage, or faster recovery. This includes actions that protect people, infrastructure, ecosystems, or services from climate impacts, even when those actions also produce economic, environmental, or emissions-related co-benefits.

If a policy measures success in terms of reduced losses, maintained service continuity, avoided disruption, or increased preparedness under climate stress, it belongs in the adaptation category. When the policy explicitly strengthens the ability to withstand or recover from climate impacts, it should be understood as resilience-focused adaptation. Adaptation policies are often place-specific and context-dependent because climate impacts vary across regions. They may involve physical interventions, institutional changes, planning processes, or social programs. Even if an adaptation policy leads to secondary benefits such as reduced emissions or economic savings, it remains classified as adaptation if its primary goal is to manage climate risk and protect people or assets from harm.

If a policy's logic begins with anticipated climate impacts and ends with increased resilience, preparedness, or safety, it should be classified as adaptation.

Operational Definition

A policy is **Adaptation** if its **primary causal pathway** is:
Climate hazard → ↓ exposure OR ↓ sensitivity OR ↑ adaptive capacity

What the model should look for

- Explicit reference to **climate impacts** (heat, flooding, drought, sea-level rise)
- **Resilience**, **preparedness**, or **risk reduction**
- Protection of people, assets, or systems from **future climate conditions**

Typical policy mechanisms

- Flood defenses
- Heat action plans
- Climate-resilient infrastructure
- Emergency response capacity
- Insurance or risk-transfer instruments

Disambiguation rule
If the policy **assumes climate change will occur and responds to its effects → Adaptation**.

Resource Efficiency

Resource efficiency policies aim to deliver the same level of service, output, or quality of life while using fewer physical resources such as energy, water, materials, or land. The core idea is that many systems are inefficient and consume more resources than necessary, and that improving efficiency can reduce costs, waste, and environmental pressure.

A policy belongs to the resource efficiency category when its primary objective is to reduce resource inputs or waste outputs per unit of activity, regardless of whether climate change is explicitly mentioned. These policies often focus on optimizing performance, improving technology, or encouraging behavioral change so that less energy, water, or material is required to achieve the same result.

Resource efficiency policies may indirectly reduce greenhouse gas emissions, especially when energy use is lowered, but emissions reduction is not the defining feature. The defining feature is improved efficiency rather than climate impact. A policy that emphasizes performance standards, efficiency benchmarks, lifecycle optimization, or waste reduction without centering its justification on climate mitigation should be classified as resource efficiency.

If a policy can be fully understood and justified even in the absence of climate change concerns, it is likely a resource efficiency policy.

Operational Definition

A policy is **Resource Efficiency** if its **primary causal pathway** is:

Same service → ↓ energy / water / material input OR ↓ waste output

What the model should look for

- Efficiency improvements **without requiring climate justification**
- Optimization of **inputs, flows, or lifecycles**
- Performance-based standards

Typical policy mechanisms

- Energy efficiency codes
- Water efficiency standards
- Circular economy policies
- Waste reduction / recycling mandates
- Industrial process optimization

Disambiguation rule

If emissions reduction is **a secondary consequence**, not the primary goal → **Resource Efficiency**.

Nature-Based Solutions

Nature-based solutions are policies that intentionally use natural systems or ecological processes to address environmental, climate, or societal challenges. The central premise is that ecosystems such as forests, wetlands, soils, and coastal systems can provide services that reduce risk, enhance resilience, or support climate goals when they are protected, restored, or managed appropriately.

A policy is classified as a nature-based solution when its effectiveness depends on the functioning of living systems rather than solely on engineered or technological infrastructure. These policies may address climate mitigation by enhancing carbon storage, adaptation by reducing exposure to hazards such as flooding or heat, or broader sustainability goals such as biodiversity protection and human well-being.

Nature-based solutions are defined by their mechanism, not their outcome. Even when a policy targets adaptation or mitigation, it should be labeled as a nature-based solution if the primary means of achieving that goal relies on ecosystems or ecological restoration. Built or hybrid systems can fall into this category when they intentionally integrate natural processes as a core component.

If removing the ecological element would fundamentally break the policy's effectiveness, the policy belongs in the nature-based solutions category.

Operational Definition

A policy is **Nature-Based Solutions** if its **primary causal pathway** is:

Ecosystem protection/restoration → climate OR resilience OR co-benefits

What the model should look for

- Explicit use of **ecosystems as infrastructure**
- Restoration, conservation, or enhancement of natural systems
- Solutions that rely on **biological processes**

Typical policy mechanisms

- Urban forests
- Wetland restoration
- Green roofs and corridors
- Mangroves, dunes, riparian buffers

Disambiguation rule

If the **mechanism depends on living systems** rather than engineered systems → **NbS**.

Cross-Category Resolution Rules

When classifying policies, priority should always be given to the policy's primary causal logic rather than secondary benefits, aspirational language, or co-benefits. A single policy may legitimately belong to more than one category, but each category should only be assigned when the underlying cause–effect relationship clearly matches the definition described above.

These rules prevent confusion and improve classification accuracy.

Rule 1: Primary mechanism > secondary benefits
Classify by **what the policy does**, not by claimed co-benefits.

Example:
"Energy efficiency to reduce emissions" → **Resource Efficiency**, not Mitigation.

Rule 2: Climate system vs climate impacts

- Acts on **emissions/carbon cycle → Mitigation**
- Acts on **hazards/vulnerability → Adaptation**

Rule 3: Natural vs engineered pathways

- Ecosystem-mediated → **NbS**
- Technological / infrastructural → other categories

Rule 4: Multi-label is allowed
Policies can legitimately be:

- Mitigation + NbS
- Adaptation + NbS
- Resource Efficiency + Mitigation, but the **primary label should be identifiable**.

Edge Cases for Adaptation Classification

1. Green roofs

- If framed as reducing urban heat exposure, managing stormwater, or protecting buildings from extreme weather → **Adaptation (Resilience)**
- If framed primarily as ecosystem restoration or urban ecology → **Nature-Based Solutions**
- If framed primarily as carbon storage → **Mitigation (secondary)**

Primary mechanism decides.

2. Passive cooling and shading

- If framed as protecting occupants during heatwaves or maintaining livable indoor temperatures during power outages → **Adaptation (Resilience)**
- If framed as reducing energy demand or emissions → **Resource Efficiency**

Classification Logic: is the goal survival and comfort under heat stress, or reduced energy consumption?